# AI: Strategy + Marketing (MGT 853)

## Introduction (Session 1)

Vineet Kumar

Yale School of Management
Spring 2026

# Agenda for Today's Session

## Course Logistics

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- What is AI? A brief history
- AI, ML, Data Science

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- AI, ML, Data Science
- Why AI now?
- What AI priorities do firms have?
- Types of ML
- AI stakeholders: consumers, firms, regulators and more

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- Evaluate the broader societal implications of AI, and how different stakeholders (consumers, employees, firms, regulators, investors and others) are impacted by AI.

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- I have included a lot of material that is optional. Why?
  - No pressure to read the optional stuff before class! Whenever you get time...

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- If you're working on your own business idea, happy to talk

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**YSOM and Course Policy**

No electronic device use (only university-approved exceptions)

# Grading & Assessment

- Grading involves both group and individual assessment

| Component | Details | Points |
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| Assignments | 1 Pairs (or Group) and 3 individual (Due Dates on Canvas) | 50 |
| Participation & Attendance | Individual | 30 |
| Project | Group | 25 |

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- What's (y)our incremental value?

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  **A: AI Business Case Development:** Develop a business case for implementing AI in a specific function to solve a specific set of problems (e.g. marketing, finance, operations).

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Your goal should be to educate an informed consumer

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- Final Project logistics (presentations) also depends on Guest Speaker availability

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**You will NOT:**

- Understand the detailed math behind how algorithms work
- Understand the best way to implement specific ML algorithms
- Develop new AI / ML algorithms and tools (Take CS courses for that)

# Course Content

# Course Outline

## Can Change Significantly! See Canvas for Latest

| Session # | Date | Topic | Assignment Due (9 am) |
|---|---|---|---|
| **Module A: AI Foundations** | | | |
| 1 | Week of Mar 23 (1) | Course Introduction and Supervised and Unsupervised Algorithms | |
| 2 | Week of Mar 23 (2) | ML Essentials | |
| 3 | Week of Mar 30 (1) | Deep Learning, Reinforcement Learning and Generative Models | **A1 (Individual / Pairs)** |
| **Module B: AI Decision Making Framework** | | | |
| 4 | Week of Mar 30 (2) | Economics of AI $\Longleftrightarrow$ Business Strategy | |
| 5 | Week of Apr 6 (1) | Decision Making with AI / Interpretable and Explainable AI | **A2 (Individual)** |
| **Module C: AI in Business + Society** | | | |
| 6 | Week of Apr 6 (2) | Zebra Medical (CASE) | |
| 7 | Week of Apr 13 (1) | Uber (CASE) | **A3 (Individual)** |
| 8 | Week of Apr 13 (2) | Borusan CAT (CASE) | |
| 9 | Week of Apr 20 (1) | Miroglio Fashion (CASE) | **A4 (Individual)** |
| 10 | Week of Apr 20 (2) | Ethical and Societal Issues | |
| **Module D: Conclusions** | | | |
| 11 & 12 | Week of Apr 27 – 30 | Speaker Presentations | **Project Due May 1** |
| 13 | Week of May 4 (1) | Course Wrap | |

# How much focus on coding?

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- ML Models Practicum (over the next couple sessions)

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- You will NOT be required to write your own code from scratch

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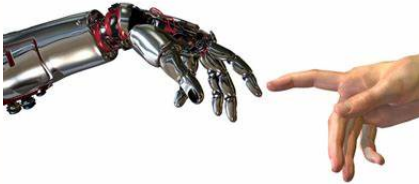

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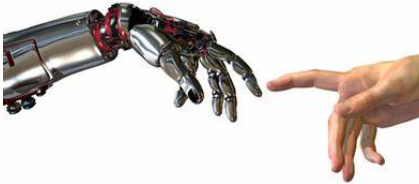

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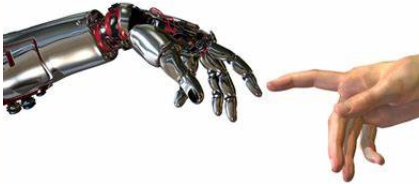

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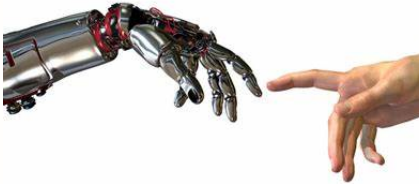

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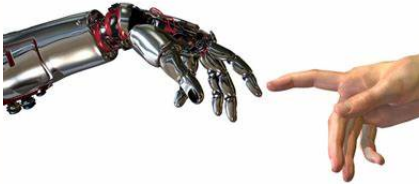

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|---|---|---|---|
| **Module A: AI Foundations** | | | |
| 1 | Week of Mar 23 (1) | Course Introduction and Supervised and Unsupervised Algorithms | |
| 2 | Week of Mar 23 (2) | ML Essentials | |
| 3 | Week of Mar 30 (1) | Deep Learning, Reinforcement Learning and Generative Models | **A1 (Individual / Pairs)** |
| **Module B: AI Decision Making Framework** | | | |
| 4 | Week of Mar 30 (2) | Economics of AI $\Longleftrightarrow$ Business Strategy | |
| 5 | Week of Apr 6 (1) | Decision Making with AI / Interpretable and Explainable AI | **A2 (Individual)** |
| **Module C: AI in Business + Society** | | | |
| 6 | Week of Apr 6 (2) | Zebra Medical (CASE) | |
| 7 | Week of Apr 13 (1) | Uber (CASE) | **A3 (Individual)** |
| 8 | Week of Apr 13 (2) | Borusan CAT (CASE) | |
| 9 | Week of Apr 20 (1) | Miroglio Fashion (CASE) | **A4 (Individual)** |
| 10 | Week of Apr 20 (2) | Ethical and Societal Issues | |
| **Module D: Conclusions** | | | |
| 11 & 12 | Week of Apr 27 – 30 | Speaker Presentations | **Project Due May 1** |
| 13 | Week of May 4 (1) | Course Wrap | |

# Historical Overview

# History of AI

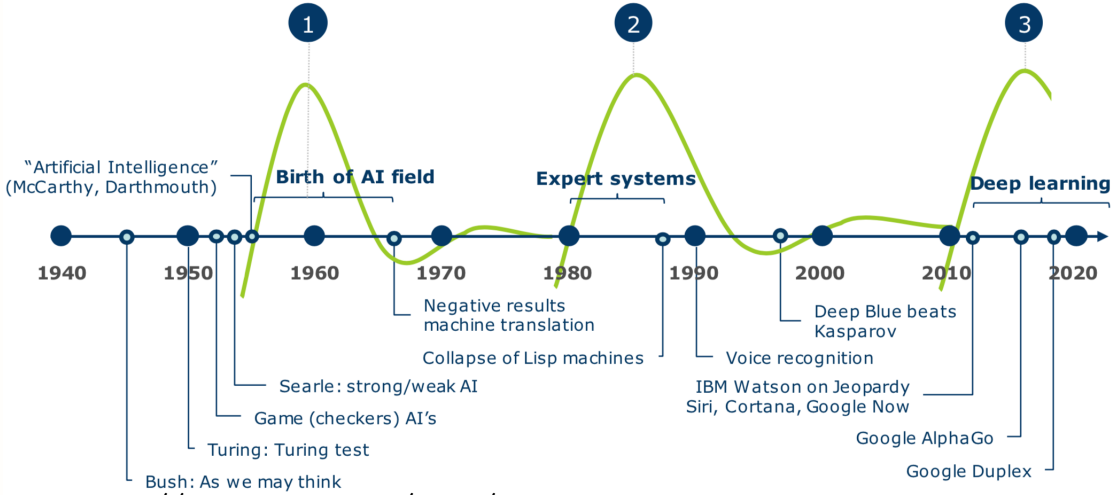
## Dartmouth College Conference

Photographer: Joe Mehling

Source: https://ojs.aaai.org/index.php/aimagazine/article/view/1911/1809

# History of AI

## Three waves of AI




Source: https://pixelspark.nl/2019/

# Humans and Machines

## Context for Turing

- Machines are very good at specific things, and can do things that humans cannot do
- A simple "machine" the wheel can go much faster than humans.
- But for a very long time, it was thought that there was one thing that humans could do that machines cannot do.
- Humans Can Think. Machines Cannot.

What exactly does it mean – "to think" ? Can Machines Think?

# Turing's View of AI

https://www.csee.umbc.edu/courses/471/papers/turing.pdf

**A. M. Turing (1950) Computing Machinery and Intelligence. *Mind 49:* 433-460.**

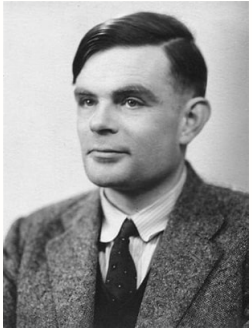
---

**COMPUTING MACHINERY AND INTELLIGENCE**

**By A. M. Turing**

**1. The Imitation Game**

I propose to consider the question, "Can machines think?" This should begin with definitions of the meaning of the terms "machine" and "think." The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous, If the meaning of the words "machine" and "think" are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning and the answer to the question, "Can machines think?" is to be sought in a statistical survey such as a Gallup poll. But this is absurd. Instead of attempting such a definition I shall replace the question by another, which is closely related to it and is expressed in relatively unambiguous words.



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In Turing's time, a parlor game:

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**Turing asks:**

If a computer (C) replaces human (M), will human interrogator (Q) know?

https://www.bbc.com/news/technology-18475646

# (Simplified) Turing's Test

**Turing suggests:**

If the computer (C) can fool the interrogator (Q) into thinking it is human, then it is said to possess **"Artificial Intelligence"**

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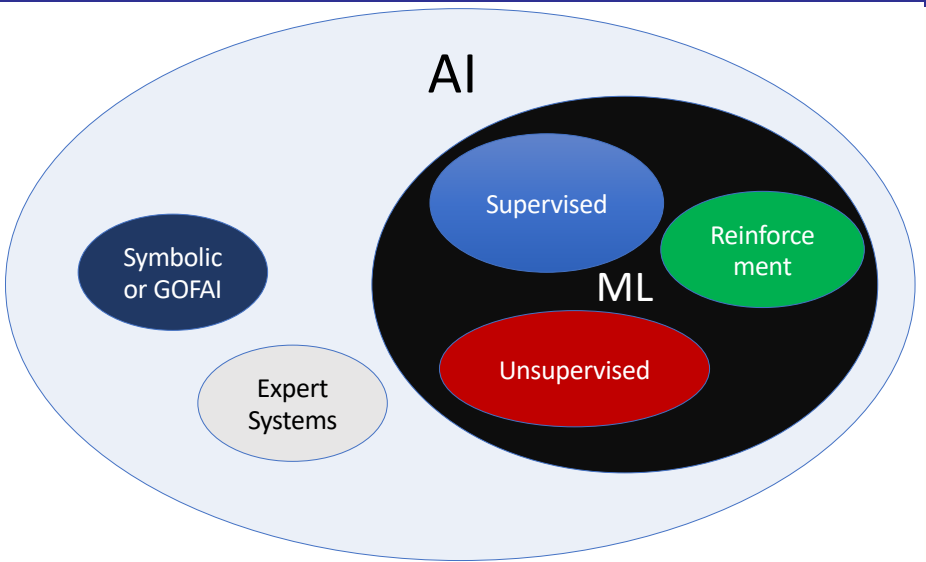
General idea for AI is to achieve human-level intelligence

# Types of AI


AI
Supervised
Reinforce
ment
ML
Unsupervised
Symbolic
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Expert
Systems

# GOFAI

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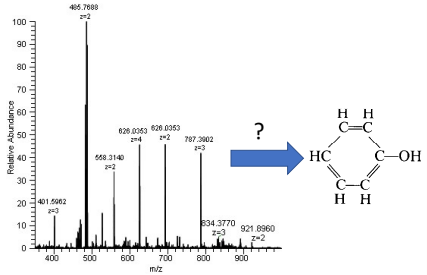
- (-) Typically suitable for small or toy problems

# Expert Systems

- "Solve problems within a specialized domain that ordinarily requires human expertise"



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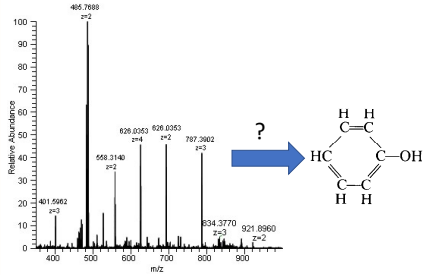

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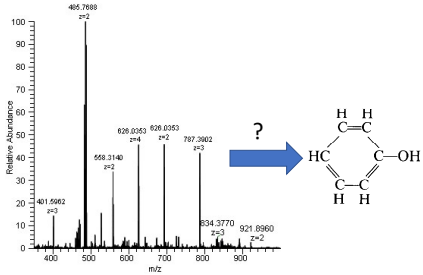

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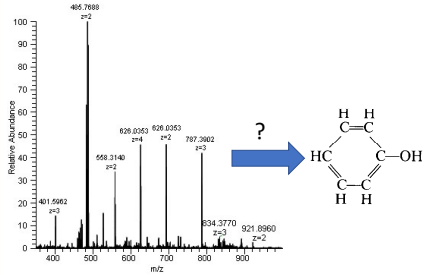

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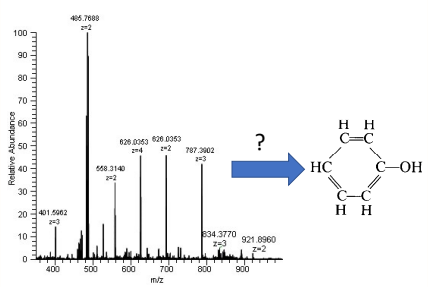
- Other applications in disease diagnosis (e.g. glaucoma), fraud detection etc.



# Definitions: AI and ML

## Artificial Intelligence

"...Intelligence can in principle be so precisely described that a machine can be made to simulate it." (John McCarthy)

## Machine Learning

- "The field of study that gives computers the ability to learn without explicitly being programmed" (Arthur Samuel)

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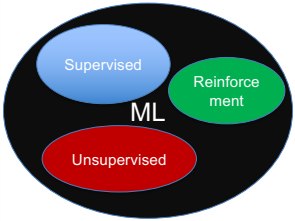
- "Improve over Task T with resect to some performance measure P based on experience E" (Tom Mitchell)

**What's the difference?**

# Differences between ML and other AI approaches

- ML primarily is "learning from data"



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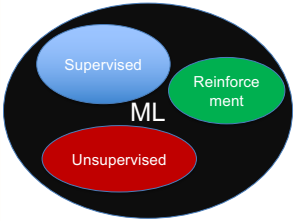

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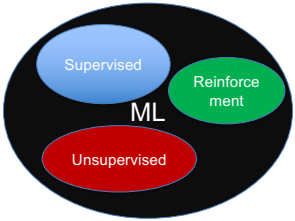

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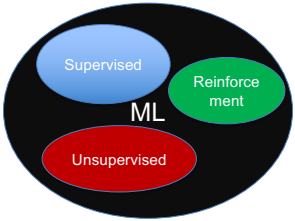

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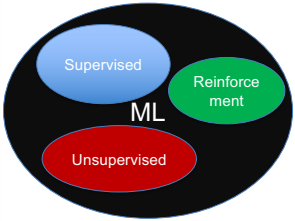

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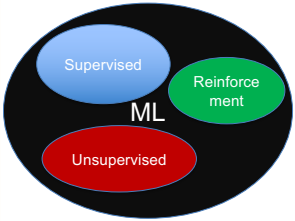


Rest of the course will focus on ML (use interchangeably with AI)

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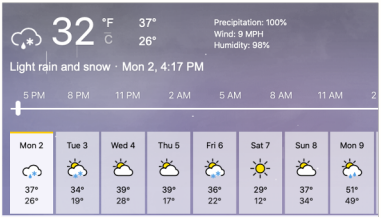
Most commonly used form of ML in practice




Logistic Regression
Polynomial Regression
Support Vector Machines
Decision Trees
Deep Neural Nets

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Can use for exploratory analysis and segmentation even when question is unclear

Cluster Analysis
K-means
K-Nearest Neighbor
Association Rule Mining
Principal Components Analysis

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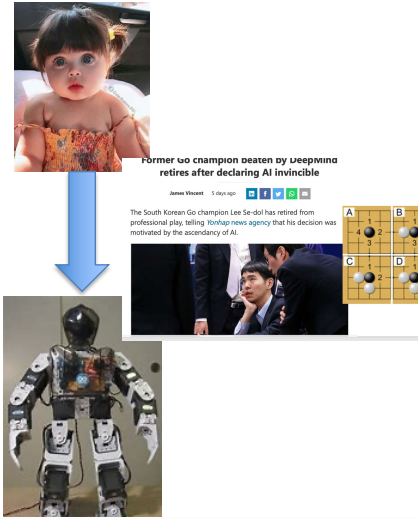
No data required



# Why AI Now?

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- Computational Advances (from GPUs)
  - What's the connection between ML and Graphics?

# Impact of AI / ML

# AI Drives Change in Firm Strategy

Google Products and Services · Artificial Intelligence · Companies · Google (company)

**What does it mean for Google to become an "AI-first" (quoting Sundar) company? How will this affect prioritization and product development?**

https://googleblog.blogspot.com/2016/04/this-years-founders-letter.html

Google sees huge value in moving from 80% accuracy in search to 99.x% accuracy

Willing to de-prioritize everything else (before this, Google was "Mobile First")

Peter Norvig, Research Director at Google

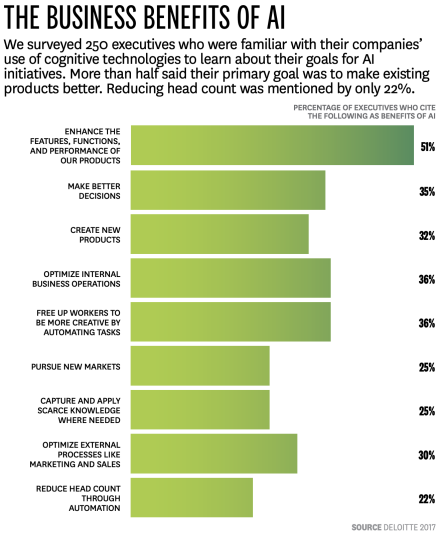
Answered May 16 2016 · Upvoted by Pål Bergerskogen, M.Sc Artificial Intelligence, Norwegian University of Science and Technology (2018) and Ken Fishkin, former Software Engineering Manager at Google (2013-2018)

"Classic" Google was an information retrieval company: you give a query, we quickly respond with ten suggestions of relevant pages, and it is your job to make sense of the suggestions. "Modern" Google, as Sundar has set out the vision, is based not just on suggestions of relevant information, but on informing and assisting. Informing, meaning that we give you the information you need, when you need it. For example, Google Now telling you it is time to leave for an appointment, or that you are now at the grocery store and previously you asked to be reminded to buy milk. And assisting means helping you to actually carry out actions—planning a trip, booking reservations; anything you can do on the internet, Google should be able to assist you in doing.

With information retrieval, anything over 80% recall and precision is pretty good—not every suggestion has to be perfect, since the user can ignore the bad suggestions. With assistance, there is a much higher barrier. You wouldn't use a service that booked the wrong reservation 20% of the time, or even 2% of the time. So an assistant needs to be much more accurate, and thus more intelligent, more aware of the situation. That's what we call "AI-first."

# What are CEO priorities in AI?

- Both external market-facing and internal
- Top external: product
- Top internal: enable employees to do higher-value jobs
- *Not so much about reducing labor costs*



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- Integrating Prediction with decision making can be challenging (Module 2)
- AI has important societal and ethical implications (Module 2)

# Next: Supervised and Unsupervised Learning

## Code

Link to Code: `https://drive.google.com/drive/folders/1L8LZvM-nEDhbUnZQzcnAv2Zo7cfwQViR`

- Familiarize yourselves with using **Google Colab** using Python:
- More practice will help you understand the next class better
- Start with the **Welcome To Colaboratory.ipynb** notebook if you're not familiar with Colab.
- **Practicum:** We will walk through code using these datasets
- Assignment 1 involves understanding and exploring the code.